Recursive Functions- Senior Division

1. **01-02 C1 Recursive Functions**

   Find $f(1)$

$$f(x)=\begin{cases} f\left(x+2\right)+x? & \text{if } x<6 \\ x-1 & \textit{otherwise} \end{cases}$$

2. **01-02 C1 Recursive Functions**

   Find $f(9,4)$

$$f(x,y)=\begin{cases} f\left(x-2,\ 2y\right)+1? & \text{if } x>6 \\ f\left(x-3,y-1\right)+2 & \text{if } 2<x\le 6 \\ 2x-y & \text{if } x\le 2 \end{cases}$$

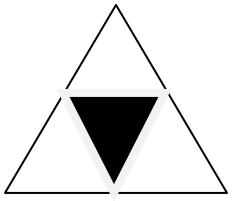
3. **02-03 C1 Recursive Functions**

   Find $f(9)$

$$f(x)=\begin{cases} f(f(x-2)+1? & \text{if } x\ge 4 \\ f\left(x-1\right)-1 & \text{if } 2\le x<4 \\ x+3 & \text{if } x<2 \end{cases}$$

4. **02-03 C1 Recursive Functions**

   Consider the following recursive algorithm: Given an equilateral triangle, connect the midpoints of the sides. Paint the triangle formed. Repeat the process 4 more times with all the remaining unpainted triangles. How many triangles will be painted?



5. **03-04 C1 Recursive Functions**

   Find $f(f(f(f(f(4)))))$

$$f(x)=\begin{cases} f\left(x-3\right)+3? & \text{if } x>3 \\ 2x+1 & \text{if } x=3 \\ x^2+2 & \text{if } x<3 \end{cases}$$

6. **03-04 C1 Recursive Functions**

Find $f(5)$

$$f(x)=\begin{cases} f(f(f(x-1)))+x & \text{if } x>3 \\ x-3 & \text{if } x\le 3 \end{cases}$$

7. **04-05 C1 Recursive Functions**

Find $f(5,1)$

$$f(x,y)=\begin{cases} f(f(f(y,x-1)),x)+1 & \text{if } x>y \\ f(x+2,y-1)-x & \text{if } x<y \\ x & \text{if } x=y \end{cases}$$

8. **04-05 C1 Recursive Functions**

Starting with a 16 inch square, bisect each side and connect the midpoints to form 4 squares inside the original square. Color the upper left and lower right hand squares. Repeat the bisection and coloring process on the two uncolored squares. Continue the process in this manner and stop when the length of a side becomes < 1. What is the total area of the colored squares?

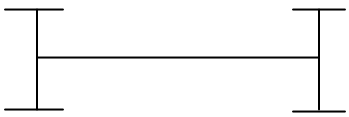
9. **05-06 C1 Recursive Functions**

Find $f(f(f(f(10))))$

$$f(x)=\begin{cases} f(x-2)-z/2 & \text{if } x\ge 5 \\ x+f(x-1) & \text{if } 2<x<5 \\ x^2\text{-}1 & \text{if } x\le 2 \end{cases}$$

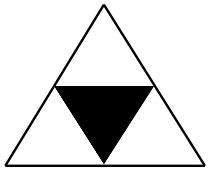
10. **05-06 C1 Recursive Functions**

Start with a segment 16 cm in length. At each end draw a segment half as long, perpendicular to the original segment and intersecting at the new segment's midpoint. If this process is continued infinitely, a fractal is created. Counting the original segment as stage 1, what is the total length of all segments after stage 6 is completed?



11. **06-07 C1 Recursive Functions**

Start with an equilateral triangle. Find the midpoint of each side.
Connect those midpoints. Remove the triangle formed.
In the second step, the procedure is applied to the 3 remaining triangles
Continuing in this manner, how many new triangles are removed in the fifth step?



12. **06-07 C1 Recursive Functions**

Find $f(100,3)$
Note that gcf(x,y) is the greatest common factor of x and y. gcf(6,8) = 2.
[x] is the largest integer function. [3.5] = 3.

$$f(x,y)=\begin{cases} \mathrm{f}\left(\left[\frac{\mathrm{x}}{\mathrm{y}}\right], y+1\right)-3 & \text{if } \mathrm{x}\geq \mathrm{y} \\ (x*y)*(\mathrm{gcf}\ (\mathrm{x},\mathrm{y})) & \text{otherwise} \end{cases}$$

13. **07-08 C1 Recursive Functions**

Find $f(12,66)$.
Note that $\mathrm{gcf}(x,y)$ is the greatest common factor of x and y. gcf(6,8) = 2.
$\mathrm{lcm}(x,y)$ is the least common multiple of x and y. lcm(6,8) = 24.

$$f(x,y)=\begin{cases} f(\mathrm{gcf}(x,y), x/\mathrm{gfc}(x,y))+\mathrm{y} & \text{if } x\geq y \\ f(\mathrm{lcm}(x,y), \mathrm{lcm}(x,y)/y)-x & \text{if } x<y \\ x^2-xy+y^2 & \text{if } x=1 \text{ or } y=1 \end{cases}$$

14. **07-08 C1 Recursive Functions**

Find $f(10)$ if: $f(1) = 2$ and $f(n+1) = 2 * f(n) + 1$

15. **08-09 C1 Recursive Functions**

Find $f(6,3)$, given:

$$f(x,y)=\begin{cases} x+f(x-1,\mathrm{y}) & \text{if } x>\mathrm{y} \\ x+f(y-2,x) & \text{if } x=\mathrm{y} \\ x+y & \text{if } x<\mathrm{y} \end{cases}$$

16. **08-09 C1 Recursive Functions**

Find $f(f(f(7)))$ given:

$$f(x)=\begin{cases} f(x-2)-1 & \text{if } x>10 \\ f(x+3)+1 & \text{if } x=10 \\ 2*x-3 & \text{if } x<10 \end{cases}$$

17. **09-10 C1 Recursive Functions**

Find $f(7)$. Express the answer as a power of 2. What is the exponent?

$$f(x)=\begin{cases} x & \text{if } x<3 \\ 2^{f(x-2)}/\text{f(x-1)} & \text{if } x\geq 3 \end{cases}$$

18. **09-10 C1 Recursive Functions**

Find $f(f(f(7)))$ given:

$$f(x)=\begin{cases} f(x-3)+x & \text{if } x>10 \\ f(x+4)-x & \text{if } x<10 \\ 0 & \text{if } x=10 \end{cases}$$

19. 1**0-11 C1 Recursive Functions**

Find $f(16)$ where [ ] is greatest integer function:

$$f(x)=\begin{cases} 2*f(x-2)+3 & \text{if } x\geq 12 \\ f([\text{x}/2])-1 & \text{if } 2\leq x<12 \\ x-4 & \text{if } x<2 \end{cases}$$

20. 1**0-11 C1 Recursive Functions**

Find $f(20)$ given:

$$f(x)=\begin{cases} f(x-6)-4 & \text{if } x\geq 10 \\ f(f(\text{x}+3))+1 & \text{if } 5<x<10 \\ 3*x & \text{if } x\leq 5 \end{cases}$$

21. 1**1-12 C1 Recursive Functions**

Find $f(16)$ given:

$$f(x)=\begin{cases} f(f(x-3))+4 & \text{if } x>8 \\ x-5 & \text{if } x\leq 8 \end{cases}$$

22. **11-12 C1 Recursive Functions**

Find f (18, 6) given:

$$f(x,y)=\begin{cases} f(x-3,y+1)-2 & if\ x>y \\ f(x+1,y) & if\ x=y \\ 3x-2y & if\ x<y \end{cases}$$

23. **12-13 C1 Recursive Functions**

Find $f(f(f(f(20))))$ given:

$$f(x)=\begin{cases} f(x/2)-1 & if\ x>0\ and\ even \\ f(x-5)-2 & if\ x>0\ and\ odd \\ x^2+3 & otherwise \end{cases}$$

24. **12-13 C1 Recursive Functions**

Find $f(2,12)$ given:

$$f(x,y)=\begin{cases} f(f(x+4,1),f(y-2,2))+x & if\ x<y \\ f(2*x,y)-y & if\ x=y \\ x+y & if\ x>y \end{cases}$$

25. **13-14 C1 Recursive Functions**

Find f (16) given:

$$f(x)=\begin{cases} f(x-3)-f(x-1) & if\ x>5 \\ 2x-1 & if\ x\le 5 \end{cases}$$

26. **13-14 C1 Recursive Functions**

Find $f(f(f(13)))$ given:

$$f(x)=\begin{cases} f(x-2)+1 & if\ x>8 \\ 2*f(x-1)-2 & if\ 5\le x\le 8 \\ x-1 & if\ x<5 \end{cases}$$

27. **14-15 C1 Recursive Functions**

Find $f(15)$ given:

$$f(x)=\begin{cases}2*f(x-3)-4 & if\ x\geq 10\\ f(x+1)+f(x-2) & if\ 8\leq x<10\\ f(x-4)-1 & if\ 5\leq x<8\\ x+3 & if\ x<5\end{cases}$$

28. **14-15 C1 Recursive Functions**

Find $f(15,12)$ given:

$$f(x,y)=\begin{cases}f(y-1,x-2)+4 & if\ x>10\\ f(x+3,y-3)+2 & if\ 5\leq x\leq 10\\ 3x-2y & if\ \text{x}\leq 4\end{cases}$$

29. **15-16 C1 Recursive Functions**

A square carpet is designed by starting with a 27 x 27 square and dividing it into 9 sections, coloring the 5 squares on the diagonals and subdividing the others into 9 sections as above. When the subdivided squares are each 1 x 1 and the diagonals are colored, how many 1 x 1 squares are uncolored?

30. **15-16 C1 Recursive Functions**

Find $f(f(f(f(16))))$, given:

$$f(x)=\begin{cases}f([x/2]-2)+3 & if\ x\geq 8\\ f(x-1)+2 & if\ 2\leq x<8\\ \text{x}+4 & if\ x<2\end{cases}$$

Note: [a] returns the greatest integer less than or equal to a